

Paul Abrams

Toronto, Ontario

Senior Full-Stack Developer · Web & Cloud · CI/CD · 15+ Years Government / Enterprise · Professional Scrum Master

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Professional Summary

Full-stack developer with **15+ years** of progressive experience building **web-based B2B and B2C applications in client/server environments**, modernizing legacy systems, and operating CI/CD pipelines at scale across the Government of Canada — Department of Justice, CIC / IRCC, ESDC (BDM). **Recurring pattern:** self-initiated automation that pays back substantial labour cost — an **IBM Rational Functional Tester** framework that eliminated a 2-week regression cycle (**~98% reduction in release validation effort, ~\$160K saved per cycle**), a self-built availability-monitoring suite that replaced volunteer evening checks, scheduled tooling that closed gaps no off-the-shelf product covered, **modernization by deletion** (decommissioned contractor-built UI-driven release apps in favour of PowerShell against REST APIs). Strong on **.NET / C#** server-side (n-tier **ASP.NET / ASP.NET MVC / ADO.NET / Entity Framework / LINQ** at 5,000+ users; **.NET Framework 2.0 / 3.5 / 4.0 / 4.5 / 5.0** and .NET 6/8 across roles), **React + TypeScript** on the front end (custom Azure DevOps web extensions, chess training app, end-to-end multiplayer web game shipped solo), and **Microsoft Azure** PaaS (Functions, Storage, SQL, APIM, Key Vault, Entra) alongside **Azure DevOps Server (TFS)** installation, configuration, customization, and multi-environment **upgrades**. Deep Microsoft toolchain: **Visual Studio 2005 → 2022, MS SQL Server 2005 / 2008 / 2008R2 / 2012 / 2016+, IIS 7 / 8 / 10, HTML / CSS / XML / XSLT / JavaScript / jQuery / AJAX, Crystal Reports / SSRS / SSIS**, OOP, **entity-relationship modelling, FDD/TDD with multi-month work-package-level effort-estimation breakdowns**. Comfortable in regulated environments — **ITSG-33 / PBMM, WCAG / AODA, ITIL. Professional Scrum Master I**. Active open-source contributor (60+ public PRs to outside repos including Microsoft's `vsts-team-calendar`, `tfs-cli`, the TypeScript-Website, MSAL .NET docs, and `canada-ca/platform-security`).

Professional Experience

Abrams Software Toronto, Ontario (Remote)

August 2025 – Present

Full-Stack Developer · Independent

- **TWNR — architected and shipped a public-facing B2C multiplayer trading game as a full-stack web application, solo.** Dockerized **TypeScript** npm monorepo: **Express + WebSocket** server backed by **PostgreSQL**, terminal-style **xterm.js** client served by Vite, individual player sign-up with **JWT auth, JSON-Schema-generated REST API contract** that doubles as live documentation. Concurrency-correct game state via PostgreSQL transactions (`SELECT ... FOR UPDATE`). Live at twnr.paulabrams.ca.
- **Chess Attack Visualization — React + TypeScript** training app consuming the **Lichess.org REST API**. Diagnosed a missing x-ray attacker feature in `chess.js` via React-lifecycle breakpoint debugging, opened [upstream PR #572](#), and published a fork as [@pabrams/chess.js](#) on npm. Live at paulabrams.ca/chess-attack-visualization.
- **Portfolio site** at paulabrams.ca operated as **Astro on GitHub Pages with GitHub Actions** CI/CD.
- **LLM code evaluation** for [DataAnnotation.tech](#) and [Stellar.ai](#) — designing prompt fixtures and grading rubrics that probe specific model failure modes across **.NET / C#, TypeScript, React, Ruby, Python, Go, Rust, and SQL**, including security audits at known-vulnerable revisions with CWE/CVSS-graded inventories, adversarial unit tests, and AI-assistant “skills” evaluated through controlled ablation.
- **60+ public PRs** to outside repos — Microsoft's `vsts-team-calendar` and `tfs-cli`, the official TypeScript-Website, the **Microsoft Authentication Library .NET** docs, `canada-ca/platform-security`, the Dygma split-keyboard configurator, and others.

Senior Software Engineer, Infrastructure and Cloud Enablement (BDM Project)

- Enhanced and maintained a **microservices** enterprise onboarding application serving **procurement, HR, and IT-ops personnel as internal customers** across the BDM project — **C# / .NET, Blazor, Azure SQL, Azure Functions, GraphQL**. Designed and consumed **REST APIs** between services; integrated with enterprise IAM via **Azure AD / Entra (OAuth 2.0, OpenID Connect)**; published and governed APIs through **Azure API Management** with authentication, throttling, and transformation policies.
- **Event-driven workflows** via **Azure Functions** — Blazor front-end submissions kicked off Functions that triggered ADO pipelines, applied ADO security and work-item updates, integrated with on-prem Active Directory via AAD Connect, and applied AAD group memberships based on personnel profile. Bulk submissions fanned out through asynchronous handoffs across the microservices.
- **Migrated the BDM Onboarding application's downstream identity** from rotating secrets (Key Vault entries + Azure DevOps PATs requiring 90-day rotation, with all auto-generated ADO work items attributed to a single person's identity) to **Azure Managed Identities** — service principals issued at runtime through Entra. Eliminated the secret-rotation incident class and properly attributed automated activity. Partnered with the Azure DevOps Services and Entra administrators on permission scoping.
- Designed and shipped **custom Azure DevOps web extensions** in **React, TypeScript, and Node.js** — a build/deploy analytics widget rendering success-rate charts via the ADO Builds REST API and Charts.js; a people-by-project reporter using an in-memory hashmap for tractable cross-referencing (the naive query pattern was unusably slow); and a weekly work-item digest emailer driven by scheduled pipelines. Responsive UI, custom styling, accessibility-aligned interactions.
- Built **PowerShell tooling** against the Azure DevOps REST API to snapshot area paths, iteration paths, security configurations, and org-level state on a schedule and commit those snapshots into Git as a **configuration-drift detector** — automation that closed a gap no off-the-shelf product covered.
- **CI/CD pipelines** in Azure DevOps for build, test, **Docker** containerization, **Kubernetes** deployment, and **Azure Data Factory** (ETL) orchestration. Implemented **ITSG-33 (NIST-based) security controls** — TLS / SSL certificate management, secret hygiene via **Azure Key Vault** — for SA&A submissions against a **PBMM** security profile.
- Administered **Azure DevOps Services** for all BDM teams; advised on custom Agile / Scrum / CMMI process templates, branching strategies, PR workflows; connected **Power BI to ADO Analytics** for richer reporting; ensured **WCAG** accessibility compliance aligned with departmental standards.
- Ran weekly **demos and design walkthroughs** for junior developers on Azure DevOps, Key Vault, app registration, and custom extensions.

Development Lead, Testing Services

- Co-architected an **internal C# / .NET web application** enabling QA staff and business analysts to author structured automated test scenarios via a web UI; designed the **SQL Server** schema and **co-designed the REST API on the web-app side** enabling bidirectional integration with Azure DevOps pipelines — script-fetch endpoints (versioned Selenium scripts generated on the fly from SQL) and results-push endpoints for structured test outcomes. Integrated with the departmental headless CMS platform.
- Engineered Azure DevOps **CI/CD pipelines for automated testing** — provisioned Windows agents to execute parallel **Selenium UI tests** against deployed applications and integrated logged test results back into ADO work items.
- Championed the departmental transition from **LoadRunner to JMeter**, reducing licensing costs while maintaining robust performance-testing capabilities. Code review and standards enforcement for the team.

Team Lead / Technical Lead

- Led **Agile / Scrum** sprint planning, review, retrospective, **daily stand-up**, and backlog refinement; pioneered capacity-planning and forecasting using Azure DevOps Server; introduced poker planning estimation across a cross-functional team.
- Administered departmental **TFS** instances and maintained heavily customized Agile, Scrum, and CMMI process templates; managed user/group security across multiple project collections.
- **Mentored developers in SQL Server query optimization** — query tuning, indexing strategy, partitioned tables, execution-plan analysis — against large TFS databases and warehouses. Also coached on SSRS and C# / .NET.
- Implemented and deployed **custom web extensions in React / TypeScript / Node.js** with responsive layouts and accessibility-aligned interactions.

Department of Justice, Canada Ottawa, ON

December 2015 – August 2017

Development Lead — iCase / Microsoft Dynamics CRM Modernization

- Dev lead on the **Legal Case Management System (iCase)** — a large n-tier **ASP.NET / C# / ADO.NET / .NET Framework 4** enterprise web application used by 5,000+ lawyers — while concurrently building its **Microsoft Dynamics CRM** successor. Modified iCase schema and front-end UI to surface migration status to users in real time, and adjusted the integration stored procedures (icisp_*) consumed by the department's SAP-based financial information system. Stack also: **Visual Studio 2013, MS SQL Server 2008R2 / 2012, IIS 7 / 8**.
- Engineered **MS Dynamics CRM plug-ins, custom actions, and workflow processes in C#**; rewrote SQL stored procedures as C# with LINQ inside CRM; used **SSIS** for ETL between iCase and Dynamics. Set up **relationships between system and custom entities**; performed **form customization with JavaScript** on CRM forms; produced **entity-relationship diagrams** and **detailed multi-month effort-estimation breakdowns (FDD/TDD)** prior to implementation. Applied design patterns (Repository, Strategy, Mediator) and clean-code refactoring.
- Established **code-review processes** integrated directly with TFS work items — gating standards for maintainability, security, and performance — and mentored junior developers.

Department of Citizenship and Immigration, Canada Ottawa, ON

April 2014 – December 2015

Senior Software Engineer · TFS / Release Engineering Lead

- Hired alongside one other engineer to take over a **TFS / TFVC** release-management environment from a contractor team. **Decommissioned over-engineered legacy C# desktop release-management apps** and replaced them with lightweight **PowerShell** scripts calling the **TFS REST APIs** — net result, most of the contractor tooling was simply deleted. **Everything about the application ended up in source control** — group memberships, permissions, repository access, process template customizations. Bias for action and ownership. Used **XSLT-via-PowerShell** to transform Work Item Template (WIT) XML for automated deployment of custom **TFS process templates**.
- **Installed, configured, customized, and upgraded TFS / Azure DevOps Server** across dev / UAT / production environments — complex multi-step upgrade paths, **Kerberos / SPN configuration** on application tiers without domain-controller access (coordinated with AD admins), database-tier migration, and process-template re-customization post-upgrade.
- Implemented and deployed Team Foundation Server **web extensions in React / TypeScript / Node.js**; maintained a fork of Microsoft's vsts-team-calendar extension with configurable colours, Outlook integration, release-management features, and IE-compatibility shims for the department's mandated browser; contributed back upstream.
- Guided development teams in migrating from legacy **XAML builds to modern, scriptable build definitions**; ran **Waterfall vs Agile** SDLC comparison sessions; wrote runbooks for TFS operation, maintenance, and troubleshooting so the work could be handed off cleanly.

Department of Justice, Canada Ottawa, ON

October 2008 – April 2014

Full-Stack Developer (2009–2014) · Software Development Engineer in Test (2008–2009)

Core contributor to the department's flagship **Legal Case Management System (iCase)** — a distributed multi-tier **ASP.NET / C#** enterprise **web-based B2B / B2C client/server** application used by 5,000+ government lawyers and contracted agent firms across Canada (agents ran their own businesses and tracked billing and timekeeping inside iCase — B2B; financial integration also crossed organizational lines within the department). Multi-tier with role- and object-based security, integrated with records management, financial, and email systems, 24/7 high availability across regionally distributed document servers. Stack: **Visual Studio 2005 / 2008 / 2010, ASP.NET / ADO.NET, .NET Framework 2.0 → 4.0, MS SQL Server 2005 → 2008R2 → 2012, IIS 6 / 7 / 8, HTML / CSS / XML / XSLT / JavaScript / jQuery / AJAX, Crystal Reports → SSRS.**

Full-Stack Developer (2009–2014)

- Led design and largely solo implementation of the **“My iCase” widget portal** — the new entry point to iCase. Ran JAD sessions, produced prototypes, authored BDD/TDD documentation. Built on **ASP.NET Web Parts and the Personalization framework** with a **custom SQL personalization provider, Knockout.js MVVM** on the front end (a direct ancestor of the **React** patterns I use today), async AJAX web-service calls, async HTTP handlers, ReportViewer in remote SSRS mode, and an HTML5 polyfill for the department-mandated IE8.
- Diagnosed a homepage report that took ~10 s uncached: profiled with **SQL Profiler, Report Execution logs, and actual execution plans**, then replaced live joins with a flat table populated by a nightly job. **~50x faster uncached, 3-4x faster cached.**
- Built a **self-initiated availability-monitoring suite** for web and document servers (an early SRE-style intervention before the term was widely used); eliminated significant overtime and reduced operational risk.
- Diagnosed and resolved a **COM Interop memory leak** across the document servers; built automated health monitoring to catch regressions.
- Co-led the **SQL Server 2005 → 2012** database upgrade; migrated core business modules from VB.NET to C#; contributed to the **Crystal Reports → SSRS** migration including 14 SSRS reports for the Legal Risk Management module.

Software Development Engineer in Test (2008–2009)

- Identified a critical regression-testing bottleneck and voluntarily architected a custom Java-based framework in **IBM Rational Functional Tester (RFT)** that parsed English-language scenario documents and auto-generated executable test scripts. Pivoted from VB.NET to Java on advice from IBM Support.
- Replaced the department's **“all-hands” manual regression cycle** (~20–25 testers running for two weeks every release) with one person reviewing automated results in five days. **~98% reduction in release validation effort**, eliminated the standing 2-week code freeze per release, **saved an estimated \$160,000 per cycle.**
- Wrote a separate test manager in C# and SQL Server. Built **round-trip code generation**: the framework could also reproduce updated English-language scenarios from the script source. Trained QA Specialists and co-op students. Contributed **500+ posts** to the IBM Developerworks Automated Functional Testing forums.

Education

Bachelor of Computer Science (Honours) - Carleton University

September 1999 - May 2003

Certifications & Awards

Professional Scrum Master I (Scrum.org) · **ITIL v3 Foundation** (Loyalist Certification Services) · **DOJ Team Merit Award 2012** (Cost Recovery Process Improvement); **Team Merit Award 2011** (Chart of Accounts); **Team Spirit Award 2009**